FINANCIAL STATEMENT ANALYSIS

(Excel + Power BI)

Submitted By

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Acknowledgment

I would like to express my gratitude to the open-source financial data platforms and resources that provided the raw corporate data necessary to build this **Financial Statement Analysis Dashboard (Excel + Power BI)** project.
This project was conceptualized, designed, and executed independently as a self-directed initiative to enhance my technical skills in financial modeling, ratio analysis, DAX measure creation, and interactive data visualization.
Working on this project independently allowed me to deepen my practical knowledge of FP&A workflows, end-to-end data processing, and dashboard architecture.

Project Overview

This project presents a comprehensive Financial Statement Analysis of Tata Consultancy Services (TCS) using Microsoft Excel and Power BI. It takes a real, publicly listed company's financial statements and converts them into an interactive dashboard covering ratio analysis, trend analysis, and management-style insights — the kind of analysis a financial or business analyst performs in a real workplace.

Objective

To analyze TCS's financial performance and position over a five-year period (FY2021–FY2025) by studying its Profit & Loss Statement and Balance Sheet, and to present the findings through an interactive, visually driven Power BI dashboard.

Scope of the Project

- Company analyzed: Tata Consultancy Services (TCS), a leading NSE/BSE-listed IT services company.
- Period covered: FY2021 to FY2025 (five financial years).
- Statements used: Consolidated Profit & Loss Statement and Consolidated Balance Sheet (IFRS).
- Data source: TCS Investor Relations – Data Sheet, tcs.com/investor-relations.

Methodology

- Collected and organized five years of raw financial data in Excel.
- Calculated key liquidity, profitability, solvency, and efficiency ratios using linked formulas.
- Performed horizontal analysis (year-on-year growth) and vertical analysis (% of revenue).
- Built an interactive Power BI dashboard with KPI cards, trend charts, and ratio visuals.
- Interpreted the results with written, analyst-style commentary on the company's financial health.

Key Deliverables

- An Excel workbook with linked raw data, ratio analysis, and trend analysis sheets.
- A Power BI dashboard visualizing five years of TCS's financial performance.
- A written summary of insights on revenue growth, profitability, liquidity, and leverage.

Tools Used

The following tools and data sources were used to complete this project:

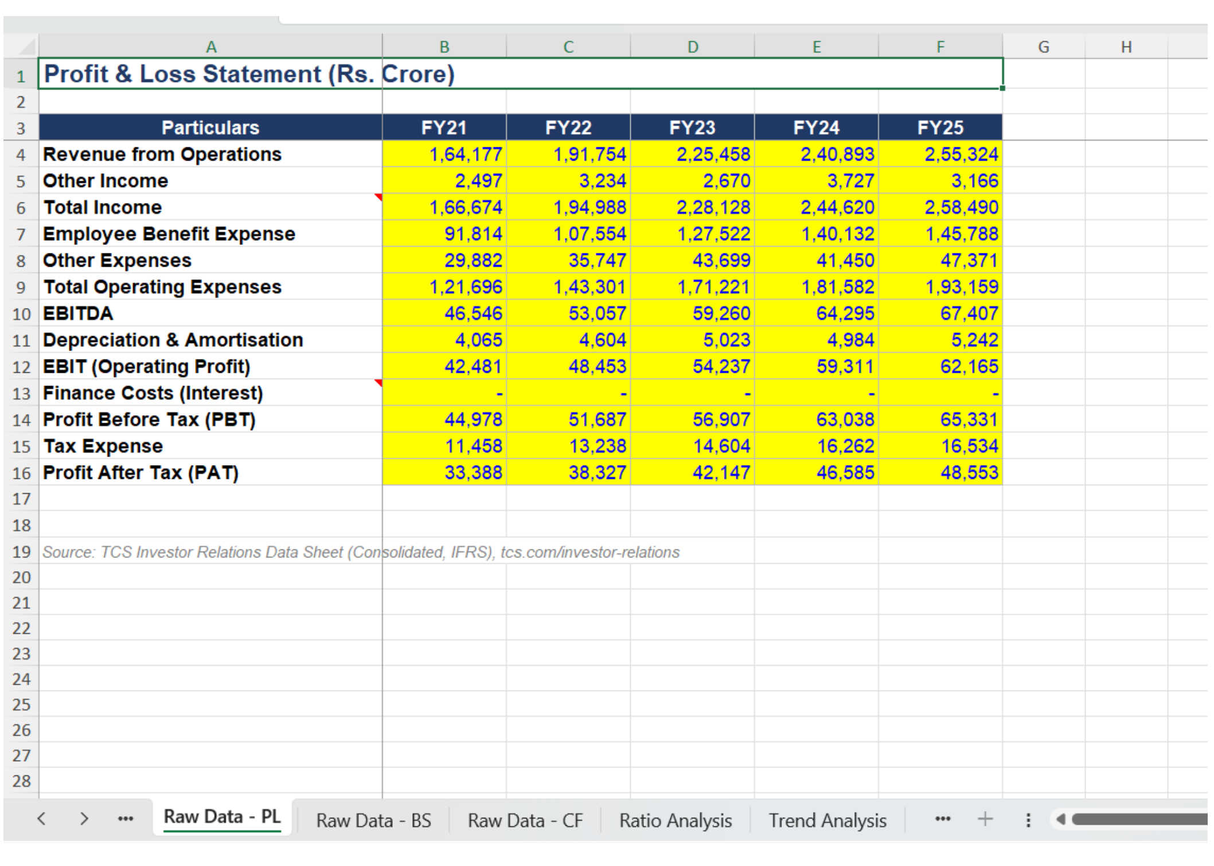
Tool	Purpose in this Project
Microsoft Excel	Data collection, formula-linked calculations for ratio analysis and trend analysis, and formatting of the raw financial data.
Microsoft Power BI	Building the interactive dashboard — KPI cards, trend charts, and ratio visualizations.
TCS Investor Relations Data Sheet	Primary source of consolidated financial statement data (FY2021–FY2025).
Microsoft Word	Preparing this project report and documentation.

Data Source & Attribution:

Dataset Link: https://github.com/afifahkhatoon61-beep/my-certificates-projects/blob/main/TCS-Data-Sheet-Q1FY27.xlsx

Financial Model & Ratio Analysis Workbook (Excel):

1.Profit & Loss Statement — Raw Data (FY21–FY25)



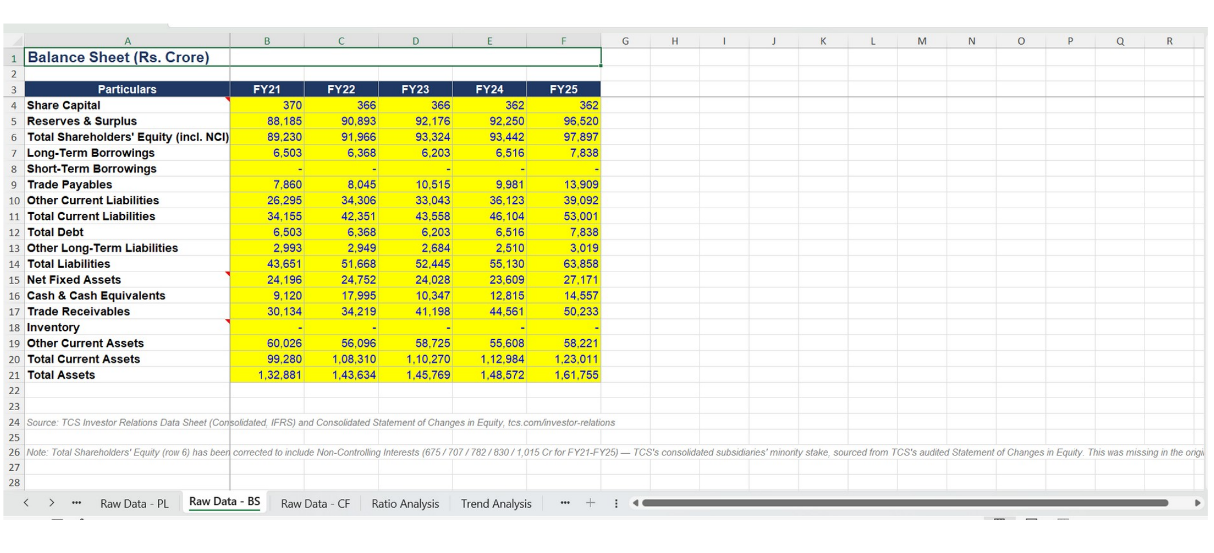
Profit & Loss Statement (Rs. Crore)

Particulars	FY21	FY22	FY23	FY24	FY25
Revenue from Operations	1,64,177	1,91,754	2,25,458	2,40,893	2,55,324
Other Income	2,497	3,234	2,670	3,727	3,166
Total Income	1,66,674	1,94,988	2,28,128	2,44,620	2,58,490
Employee Benefit Expense	91,814	1,07,554	1,27,522	1,40,132	1,45,788
Other Expenses	29,882	35,747	43,699	41,450	47,371
Total Operating Expenses	1,21,696	1,43,301	1,71,221	1,81,582	1,93,159
EBITDA	46,546	53,057	59,260	64,295	67,407
Depreciation & Amortisation	4,065	4,604	5,023	4,984	5,242
EBIT (Operating Profit)	42,481	48,453	54,237	59,311	62,165
Finance Costs (Interest)	-	-	-	-	-
Profit Before Tax (PBT)	44,978	51,687	56,907	63,038	65,331
Tax Expense	11,458	13,238	14,604	16,262	16,534
Profit After Tax (PAT)	33,388	38,327	42,147	46,585	48,553

Source: TCS Investor Relations Data Sheet (Consolidated, IFRS), tcs.com/investor-relations

explanation:
This sheet consolidates TCS's five-year Profit & Loss data (FY21–FY25) in Rs. Crore, sourced from TCS's Investor Relations Data Sheet. All key line items — Revenue, Operating Expenses, EBITDA, PBT, and PAT — feed directly into the Ratio Analysis and Trend Analysis sheets via linked formulas. Note: Depreciation & Amortisation is shown as its own line (for EBITDA calculation) but is already embedded within "Other Expenses" as reported by TCS — so Total Operating Expenses (₹1,21,696 Cr, FY21) reflects the full cost base including D&A, and EBIT = Revenue − Total Operating Expenses, while EBITDA = EBIT + D&A added back.

2.Balance Sheet — Raw Data (FY21–FY25):



Balance Sheet (Rs. Crore)

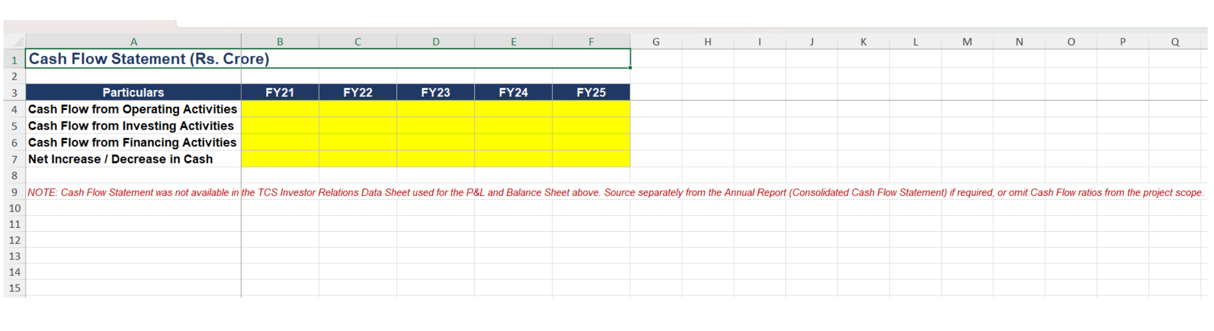
Particulars	FY21	FY22	FY23	FY24	FY25
Share Capital	370	366	366	362	362
Reserves & Surplus	88,185	90,893	92,176	92,250	96,520
Total Shareholders' Equity (incl. NCI)	89,230	91,966	93,324	93,442	97,897
Long-Term Borrowings	6,503	6,368	6,203	6,516	7,838
Short-Term Borrowings	-	-	-	-	-
Trade Payables	7,860	8,045	10,515	9,981	13,909
Other Current Liabilities	26,295	34,306	33,043	36,123	39,092
Total Current Liabilities	34,155	42,351	43,558	46,104	53,001
Total Debt	6,503	6,368	6,203	6,516	7,838
Other Long-Term Liabilities	2,993	2,949	2,684	2,510	3,019
Total Liabilities	43,651	51,668	52,445	55,130	63,858
Net Fixed Assets	24,196	24,752	24,028	23,609	27,171
Cash & Cash Equivalents	9,120	17,995	10,347	12,815	14,557
Trade Receivables	30,134	34,219	41,198	44,561	50,233
Inventory	-	-	-	-	-
Other Current Assets	60,026	56,096	58,725	55,608	58,221
Total Current Assets	99,280	1,08,310	1,10,270	1,12,984	1,23,011
Total Assets	1,32,881	1,43,634	1,45,769	1,48,572	1,61,755

Source: TCS Investor Relations Data Sheet (Consolidated, IFRS) and Consolidated Statement of Changes in Equity, tcs.com/investor-relations

Note: Total Shareholders' Equity (row 6) has been corrected to include Non-Controlling Interests (675 / 707 / 782 / 830 / 1,015 Cr for FY21-FY25) — TCS's consolidated subsidiaries' minority stake, sourced from TCS's audited Statement of Changes in Equity. This was missing in the origi

explanation:
This sheet captures TCS's five-year Balance Sheet data (FY21–FY25) in Rs. Crore, covering equity, liabilities, and assets, sourced from TCS's Investor Relations Data Sheet. Total Shareholders' Equity includes Non-Controlling Interests (₹675–1,015 Cr across FY21–FY25) — the minority shareholders' stake in TCS's consolidated subsidiaries — ensuring Total Assets equals Total Equity plus Total Liabilities in every year. These figures power the liquidity, solvency, and efficiency ratios calculated automatically in the Ratio Analysis sheet.

3.Cash Flow Statement — Out of Scope



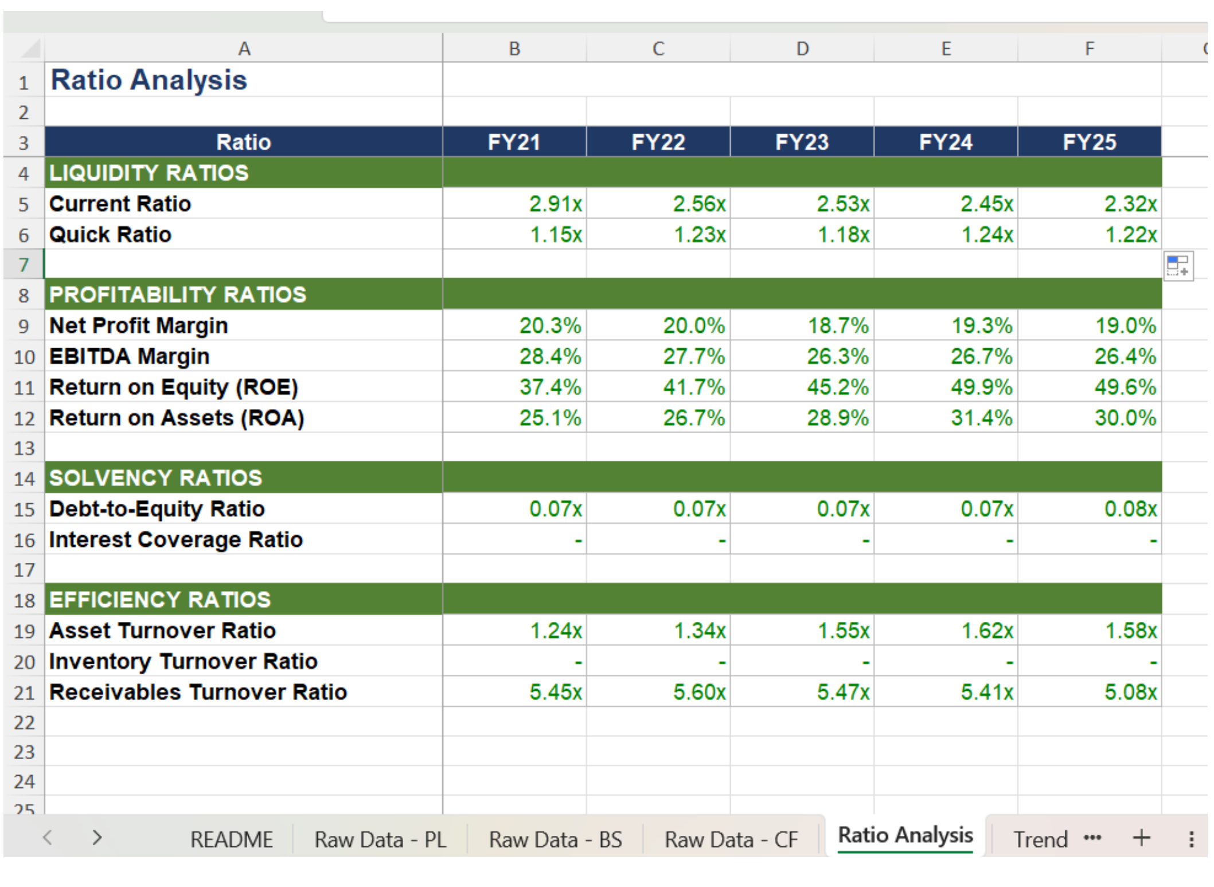
Cash Flow Statement (Rs. Crore)

Particulars	FY21	FY22	FY23	FY24	FY25
Cash Flow from Operating Activities					
Cash Flow from Investing Activities					
Cash Flow from Financing Activities					
Net Increase / Decrease in Cash					

NOTE: Cash Flow Statement was not available in the TCS Investor Relations Data Sheet used for the P&L and Balance Sheet above. Source separately from the Annual Report (Consolidated Cash Flow Statement) if required, or omit Cash Flow ratios from the project scope.

explanation:
Cash Flow data was not available in TCS's Investor Relations Data Sheet used for the P&L and Balance Sheet analysis above, and was therefore excluded from the project scope. The ratio and trend analysis in this project is based entirely on the Profit & Loss Statement and Balance Sheet.

Ratio Analysis — Liquidity, Profitability, Solvency & Efficiency (FY21–FY25)

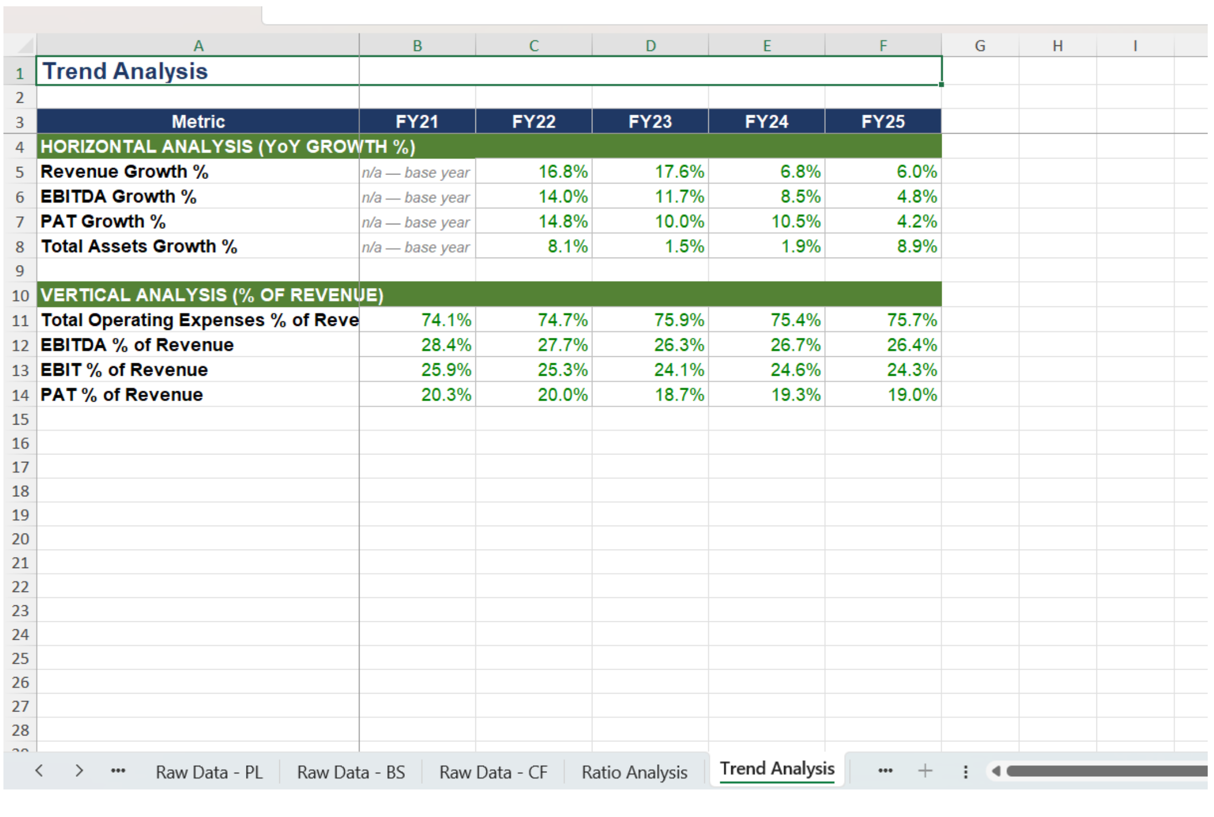


Ratio Analysis

Ratio	FY21	FY22	FY23	FY24	FY25
LIQUIDITY RATIOS					
Current Ratio	2.91x	2.56x	2.53x	2.45x	2.32x
Quick Ratio	1.15x	1.23x	1.18x	1.24x	1.22x
PROFITABILITY RATIOS					
Net Profit Margin	20.3%	20.0%	18.7%	19.3%	19.0%
EBITDA Margin	28.4%	27.7%	26.3%	26.7%	26.4%
Return on Equity (ROE)	37.4%	41.7%	45.2%	49.9%	49.6%
Return on Assets (ROA)	25.1%	26.7%	28.9%	31.4%	30.0%
SOLVENCY RATIOS					
Debt-to-Equity Ratio	0.07x	0.07x	0.07x	0.07x	0.08x
Interest Coverage Ratio	-	-	-	-	-
EFFICIENCY RATIOS					
Asset Turnover Ratio	1.24x	1.34x	1.55x	1.62x	1.58x
Inventory Turnover Ratio	-	-	-	-	-
Receivables Turnover Ratio	5.45x	5.60x	5.47x	5.41x	5.08x

explanation:
TCS shows strong liquidity (Current Ratio 2.32x–2.91x, Quick Ratio 1.15x–1.22x), excellent profitability (Net Profit Margin ~19-20%, ROE improving from 37.4% to 49.6% after including NCI in Total Equity), minimal leverage (Debt-to-Equity below 0.1x), and efficient asset use (Asset Turnover up from 1.24x to 1.58x). Interest Coverage and Inventory Turnover are not applicable, as TCS carries negligible debt and holds no physical inventory. All ratios are formula-linked to raw data — nothing hardcoded.

Trend Analysis — Horizontal & Vertical (FY21–FY25):

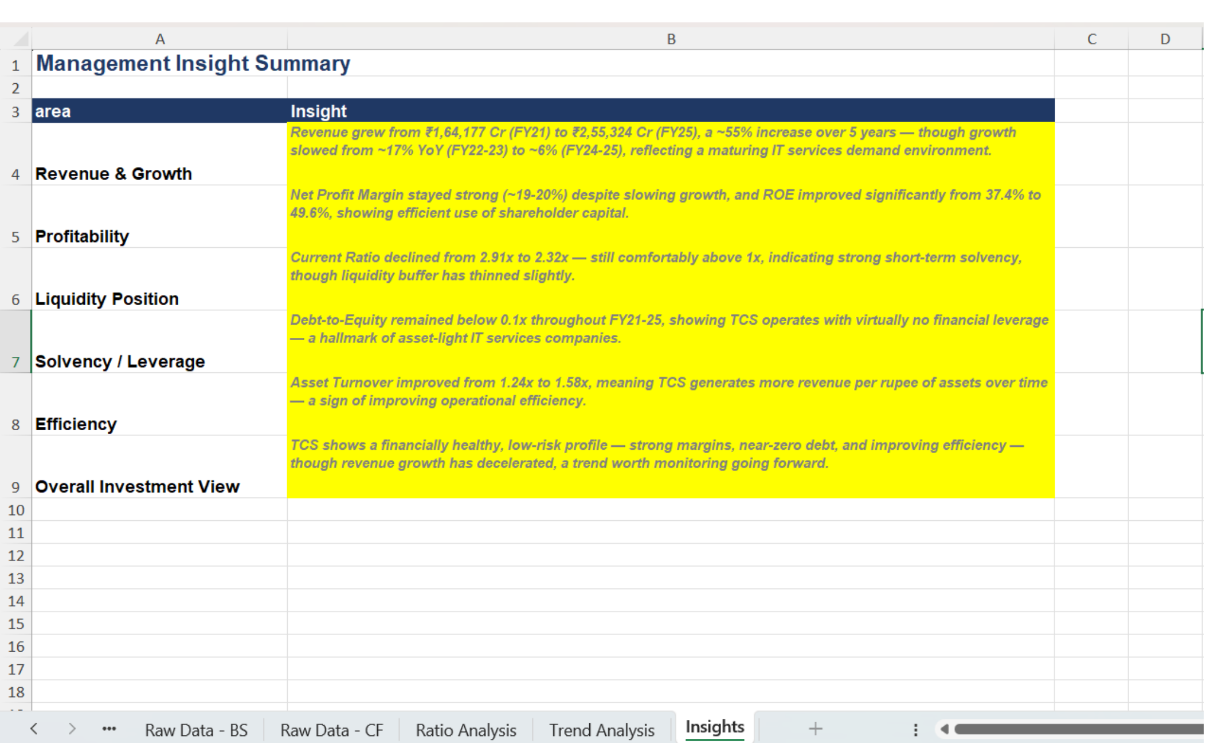


Trend Analysis

Metric	FY21	FY22	FY23	FY24	FY25
HORIZONTAL ANALYSIS (YoY GROWTH %)					
Revenue Growth %	n/a — base year	16.8%	17.6%	6.8%	6.0%
EBITDA Growth %	n/a — base year	14.0%	11.7%	8.5%	4.8%
PAT Growth %	n/a — base year	14.8%	10.0%	10.5%	4.2%
Total Assets Growth %	n/a — base year	8.1%	1.5%	1.9%	8.9%
VERTICAL ANALYSIS (% OF REVENUE)					
Total Operating Expenses % of Reve	74.1%	74.7%	75.9%	75.4%	75.7%
EBITDA % of Revenue	28.4%	27.7%	26.3%	26.7%	26.4%
EBIT % of Revenue	25.9%	25.3%	24.1%	24.6%	24.3%
PAT % of Revenue	20.3%	20.0%	18.7%	19.3%	19.0%

explanation:
Horizontal analysis tracks year-on-year growth in Revenue, EBITDA, and PAT, while vertical analysis expresses each cost and profit line as a percentage of revenue. Together, these reveal that revenue growth slowed from ~17% (FY22–23) to ~6% (FY24–25), while the cost structure remained stable — Total Operating Expenses held steady around 74–76% of revenue throughout. Note: Total Operating Expenses already includes Depreciation & Amortisation (see P&L sheet note), so Total Opex % + EBIT % of Revenue sums to exactly 100% each year.

Management Insight Summary — Analyst Commentary



Management Insight Summary

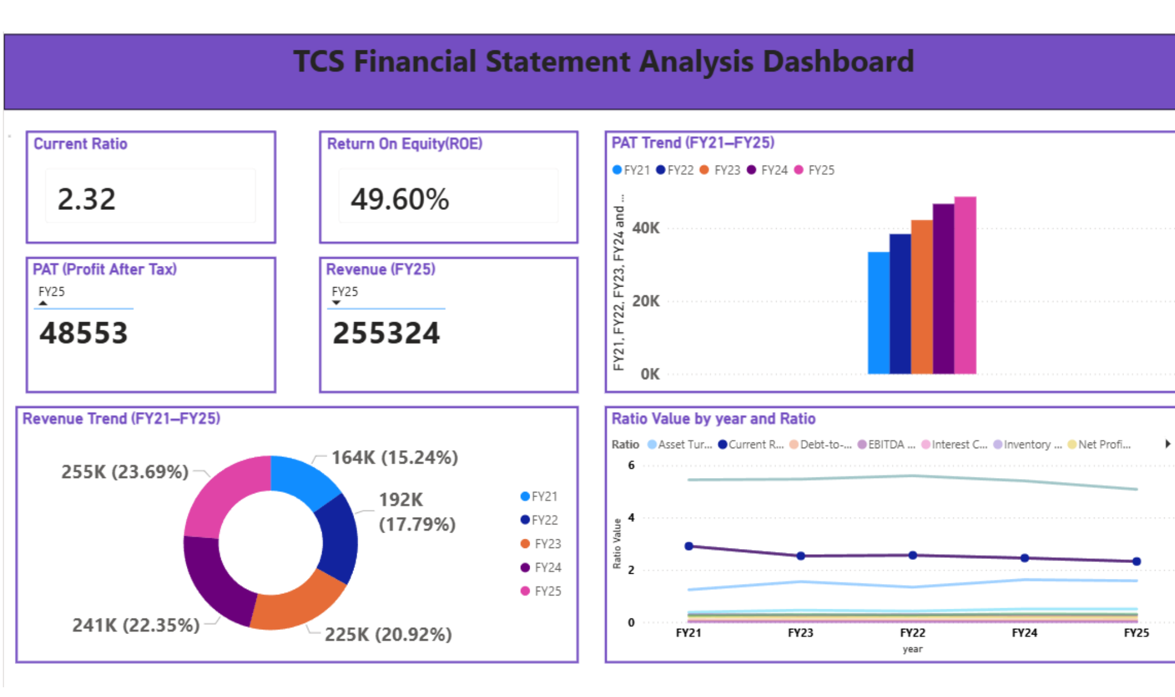
area	Insight
Revenue & Growth	*Revenue grew from ₹1,64,177 Cr (FY21) to ₹2,55,324 Cr (FY25), a ~55% increase over 5 years — though growth slowed from ~17% YoY (FY22-23) to ~6% (FY24-25), reflecting a maturing IT services demand environment.*
Profitability	*Net Profit Margin stayed strong (~19-20%) despite slowing growth, and ROE improved significantly from 37.4% to 49.6%, showing efficient use of shareholder capital.*
Liquidity Position	*Current Ratio declined from 2.91x to 2.32x — still comfortably above 1x, indicating strong short-term solvency, though liquidity buffer has thinned slightly.*
Solvency / Leverage	*Debt-to-Equity remained below 0.1x throughout FY21-25, showing TCS operates with virtually no financial leverage — a hallmark of asset-light IT services companies.*
Efficiency	*Asset Turnover improved from 1.24x to 1.58x, meaning TCS generates more revenue per rupee of assets over time — a sign of improving operational efficiency.*
Overall Investment View	*TCS shows a financially healthy, low-risk profile — strong margins, near-zero debt, and improving efficiency — though revenue growth has decelerated, a trend worth monitoring going forward.*

explanation:
This section translates the raw ratios and trends into analyst-style written insights covering revenue growth, profitability, liquidity, leverage, and efficiency — concluding with an overall assessment of TCS's financial health based on the FY21–FY25 analysis, using the corrected (NCI-inclusive) equity figures.

Project Deliverable (Excel Workbook):

https://github.com/afifahkhatoon61-beep/my-certificates-projects/blob/main/FSA_Dashboard_TCS_filled__in%20ms%20excel.xlsx

Power BI Dashboard:



Power BI Dashboard — Key Points:

- Interactive dashboard visualizing TCS's consolidated financial performance across FY21–FY25, built on top of the Excel-based Ratio and Trend Analysis model
- **KPI Summary Cards**: Current Ratio (2.32x), Return on Equity (49.6%), PAT (₹48,553 Cr), and Revenue (₹2,55,324 Cr) for FY25 — instant snapshot of latest financial health
- **PAT Trend Chart**: Column chart showing consistent year-on-year profit growth from FY21 to FY25
- **Revenue Trend Chart**: Donut chart breaking down each year's share of total 5-year revenue (15.24% in FY21 → 23.69% in FY25)
- **Ratio Value Chart**: Multi-line view consolidating all liquidity, profitability, solvency, and efficiency ratios in one place for easy trend comparison
- **Data Connectivity**: Built via Power Query on the corrected Excel workbook, so it automatically reflects the reconciled Total Equity (including Non-Controlling Interests) — keeping the dashboard accurate and audit-ready

Power BI Dashboard File:

https://github.com/afifahkhatoon61-beep/my-certificates-projects/blob/main/tcs%20financial%20statement%20analysis%20dashboard%20in%20power%20Bi%20file.pbix

Key takeaways from the analysis:

- **Strong and Improving Profitability**: ROE improved from 37.4% (FY21) to 49.6% (FY25), and Net Profit Margin remained consistently strong at 18.7%–20.3%, reflecting efficient conversion of revenue into profit
- **Steady Revenue & Profit Growth**: Revenue grew from ₹1.64L Cr to ₹2.55L Cr, and PAT grew from ₹33,388 Cr to ₹48,553 Cr over FY21–FY25
- **Minimal Financial Leverage**: Debt-to-Equity Ratio stayed below 0.1x throughout the period, indicating TCS relies almost entirely on internal accruals rather than external debt — typical of an asset-light IT services company
- **Healthy Liquidity Position**: Current Ratio (2.32x–2.91x) and Quick Ratio (1.15x–1.22x) confirm TCS comfortably covers short-term obligations, even under a conservative liquidity test
- **Efficient Asset Utilization**: Asset Turnover Ratio improved from 1.24x to 1.58x, and Receivables Turnover stayed healthy (5.08x–5.60x), showing effective use of assets and disciplined collections
- **Data Integrity Validated**: Balance Sheet reconciled exactly (Assets = Equity + Liabilities) after correctly including Non-Controlling Interests in Total Equity — a real audit finding from the analysis, not just a cosmetic fix

Conclusion:

This five-year financial statement analysis of Tata Consultancy Services (FY21–FY25) confirms a company in strong financial health — consistent revenue growth, improving profitability and capital efficiency, near-zero financial leverage, and a comfortable liquidity position, all typical of a mature, asset-light IT services leader. While topline growth has moderated in the last two years, TCS's ability to expand margins and returns on equity alongside this slowdown reflects disciplined cost management and efficient capital allocation.

Beyond the financial findings, this project also involved a full reconciliation of the underlying data against TCS's own audited financial statements — identifying and correcting a Balance Sheet discrepancy (a missing Non-Controlling Interests component in Total Equity) and validating every figure in the Profit & Loss, Balance Sheet, and Ratio Analysis sheets line-by-line. This end-to-end process, from raw data sourcing to error detection, correction, and dashboard integration, reflects an accuracy-first, detail-oriented approach to financial analysis.

"THANK YOU"